Since only one equilateral triangle could be inscribed in or circumscribed around a given circle.

Needs proof

Author: Very good. I see that you have grasped the essence of the concept of functional relationship. I should emphasize that from the broadest point of view this concept is based on the idea of mapping one set of objects onto another set of objects. It is obvious that a list of different functions may be unlimited.

- And now we shall examine the already familiar concept of a numerical sequence as an example of mapping.

Reader: A numerical sequence is, apparently, a mapping of a set of natural numbers onto a different numerical set. The elements of the second set are the terms of the sequence. Hence, a numerical sequence is a particular case of a numerical function. The domain of a function is represented by a set of natural numbers.

Author: This is correct. But you should bear in mind that later on we shall deal with numerical functions whose domain is represented by the real line, or by its interval (or intervals), and whenever we mention a function, we shall imply a numerical function.

In this connection it is worthwhile to remind you of the classification of intervals. In the previous dialogue we have already used this classification, if only partially.

First of all we should distinguish between the intervals of finite length:

- a closed interval that begins at a and ends at b is denoted by $[a,b]$; the numbers x composing this interval meet the inequalities $a \leq x \leq b$;
- an open interval that begins at a and ends at b is denoted by $]a,b[$; the numbers x composing this interval meet the inequalities $a < x < b$;
- a half-open interval is denoted either by $]a,b]$ or $[a,b[$, the former implies that $a < x \leq b$, and the latter that $a \leq x < b$.

The intervals may also be infinite:

$]-\infty,\infty[$ $(-\infty < x < \infty)$ — the real line

$]a,\infty[$ $(a < x < \infty)$; $[a,\infty[$ $(a \leq x < \infty)$